

# Leapfrogging like never before

**E**very year, towards the end of the calendar, our test labs across the country turn into performance war rooms. Rows of laptops, desktops, phones, routers and assorted gadgets queue up for benchmarking, all chasing the “best of 2025” crown.

Across this season’s Zero1 Awards testing cycle, one pattern keeps showing up. New laptop processors are not just a little faster than last year’s stars, they are storming ahead by hundreds or even thousands of points in synthetic benchmarks. Desktop graphics cards are doing the same, turning once-impossible 4K settings into the new default. On the AI side, every few months seems to bring a foundational model that makes the previous “state of the art” feel modest.

It used to be that benchmark designers could rely on a comfortable headroom. You could craft a workload just tough enough to make top-tier hardware sweat and expect it to last a few years. Now it feels like the tests are the ones being outpaced. Each cycle demands more punishing, more parallel, more memory-hungry workloads just to expose the differences between flagships. The race is no longer just product versus product, it is also benchmarks versus time.

The irony is that while scores soar, many people’s workloads have not become proportionally more demanding. For them, the last few years of performance gains mainly translate into everything feeling instant, then more instant, then indistinguishably instant.

That is where the hero features come in. The obvious one is the NPU, the dedicated neural processing block quietly spreading through modern laptops and phones. NPUs are improving, but not quite at the breakneck pace of general CPU and GPU performance. Their presence is less about raw numbers and more about future promises: local copilots, live translation, on-device media enhancements, smarter background processing. Yes, these aren’t exactly ‘future promises’ but if you’ve used any of these features, they’re always faltering. That aside, these features give marketing teams a clear narrative, even



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**Mithun Mohandas**  
Managing Editor



Let me know your thoughts on this column at:  
**@mithun\_mohandas | mithun@digit.in**

if the average user is not yet saturating those capabilities for now.

So what happens when the hardware curve outgrows everyday needs?

One possibility is that upgrade cycles stretch even further. If a three-year-old machine already feels excessive for basic tasks, the incentive to buy new shrinks, at least on the consumer side. Enthusiasts will still chase higher frame rates and creators will still value shorter render times, but that is not the bulk of the market.

From the industry’s perspective, this is where AI infrastructure starts to look like the real lifeline. The same companies that ship consumer CPUs, GPUs and NPUs are also building data centre accelerators by the truckload. Hyperscalers and enterprises are hungry for compute to train and serve ever larger models. That demand is not tied to someone debating an upgrade for their home laptop, it is tied to competitive pressure in cloud, search, enterprise software and national AI programmes.

In such a world, consumer hardware risks becoming a side effect of the AI arms race. Architectural innovations developed for massive accelerators trickle down into gaming GPUs and laptop chips. Retail products remain astonishingly powerful, but their sales are no longer the sole pillar holding up quarterly earnings.

There is also a softer shift under way, one that has more to do with positioning than with silicon. If performance has effectively cleared the bar for mainstream users, companies will need to lean harder on experience, software, ecosystems and sustainability. Longer support windows, better drivers, calmer thermals, quieter fans, and thoughtful AI features might convince buyers more than another 20% on a synthetic chart.

The strange thing about this era is that the leapfrogging is real, measurable and impressive, but quietly eroding urgency to keep up. When even the “old” kit feels fast, the upgrade pitch changes. The arms race carries on at full tilt in the cloud, not in the living room. **d**